Inwoo Lee

https://github.com/InwooLeeme
solved.ac/profile/inwooleeme
[Web page]

Department of Software Engineering, Jeonbuk National University

Education

- **Jeonbuk National University** *Mar. 2019 - Aug. 2026*
  B.S. in Software Engineering GPA: 3.69/4.5
- **SKALA (SK AI Leader Academy), 4th Cohort** *Jul. 2026 - Dec. 2026 (expected)*
  AI talent development program by SK Group

Extracurricular Activity

- **JBNU Programming Contest (JBNUPC)** *2024 - 2026*
  Problem Setter & Contest Staff (ALPS, JBNU SW-Centered University TA)
  - Authored contest problems for JBNUPC 2024 and 2025, and set and operated JBNUPC 2026
  - Reviewed statements, edge cases, and test data against every published input constraint
- **SKOJ - Online Judge for SKALA Trainees** *2026*
  Co-developer & Problem Setter
  - Authored about 350 problems, covering statements, reference solutions, test data, and review
  - Automated problem generation and upload with an AI-assisted workflow, roughly doubling throughput
- **ALPS - Algorithm Study Club, Jeonbuk National University** *Mar. 2024 - Dec. 2025*
  Member
  - Solved and shared weekly problem sets; UCPC qualifier 132nd (2024) and 135th (2025)

Achievements

- **Gold Prize** 2025 JBNU ICPC Contest (team of 3) *Oct. 2025*
- **Excellent Paper Award** KIIT Undergraduate Paper Competition (Bronze) *Jun. 2025*
- **Gold Prize** 2024 JBNU ICPC Contest (team of 3) *Nov. 2024*
- **Encouragement Award** 7th SW Programming Contest, Honam & Jeju Region *Oct. 2024*

Projects

- **MCP Assistant Voice-Controlled PC Assistant with MCP** *Mar. 2025 - Jun. 2025*
  Capstone project (team of 4), published at KIIT 2025; rebuilt solo in 2026
  - Built browser and notepad MCP tools and the agent's tool calling so commands run end to end
  - Merged a 3-stage agent pipeline into one planner: LLM calls per command 3 → 1, latency ~2s → 0.5s
- **Molecular Toxicity Prediction** *2026*
  Classifies molecular toxicity from structure and property data (team of 6)
  - Compared features and models (RF, ExtraTrees, XGBoost, LightGBM) and built the final notebooks
  - Scaffold-based train/validation split to prevent leakage; RF + ExtraTrees ensemble reached F1 0.8483
- **NASDAQ Pattern Chart** *2024 - 2026*
  Candlestick chart and similar-pattern search over 11,125 daily bars (database course, team of 4)
  - Gzip-compressed API responses: payload 1.99MB → 263KB, latency ~2s → 0.15-0.49s
  - Replaced SVG-based ApexCharts with canvas-based lightweight-charts: JS bundle 183KB → 99KB

Skills

- **Technical skills:** C++, Python, TypeScript, Java, FastAPI, Spring, Next.js, React, LangGraph, AutoGen, MCP, LLM, Docker, GitHub Actions, AWS EKS, Argo CD
- **Certifications:** Engineer Information Processing (2025), ADsP (2026), Korean History Test Level 1 (2025)